Scale-Conditioned Geometry of the Wildland–Urban Interface: A Theory of Anthropogenic–Ecological Boundaries

First Author[1,*] and Second Author[1]

[1]Department of Earth and Environmental Sciences, University Name, City, State ZIP, USA
*** Correspondence:** author@institution.edu
Manuscript Type: Article

Abstract

Boundary length is widely used to characterize exposure in the wildland–urban interface (WUI), yet it is often reported as a context-free scalar as if it were an intrinsic property of the landscape. This practice overlooks a fundamental geometric reality: the length of any mapped boundary depends both on how the boundary is defined and on the scale at which it is measured. Here we develop a scale-conditioned framework for WUI boundary inference that distinguishes two independent sources of variation. The first is *definition scale*, which determines the boundary object through classification design choices such as settlement representation, vegetation thresholds, neighborhood radius, and adjacency rules. The second is *measurement scale*, which determines the estimated perimeter of that object through geometric resolution. Drawing on WUI mapping research together with scaling theory from fractal geometry and wildfire perimeter analysis, we formalize WUI boundary length as a family of functions $L_d(\varepsilon)$, where d indexes defensible definition bundles and ε indexes measurement scale. We then extend this framework by interpreting the WUI boundary as a generative anthropogenic–ecological interface produced by interacting spatial processes: infrastructure diffusion, vegetation mosaic dynamics, and wildfire disturbance. In this view, WUI boundary geometry encodes information about coupled settlement expansion and ecological pattern formation. The principal result is that apparent disagreements in reported WUI boundary lengths are often expected consequences of scale specification rather than contradictory empirical evidence. Recasting WUI boundary length as a scale-conditioned estimand therefore provides a conceptual foundation for transparent wildfire risk analysis, land-use planning, insurance analytics, and cross-jurisdictional comparison while situating WUI within a broader class of anthropogenic edge systems.

Keywords: wildland–urban interface; wildfire; boundary scaling; fractal geometry; anthropogenic–ecological systems; geospatial inference

1. Introduction

The wildland–urban interface (WUI) represents one of the most consequential spatial configurations in wildfire science. Where settlements intermix with flammable vegetation, fire behavior intersects directly with the built environment, creating coupled socio-ecological risk systems that influence land management, infrastructure planning, insurance markets, and public safety. As development expands into vegetated landscapes, the spatial extent and geometry of WUI have become central concerns for mitigation and climate adaptation (Cohen, 2000, Radeloff et al., 2005, Carlson et al., 2022, Radeloff et al., 2018, Bar-Massada et al., 2014, Calkin et al., 2014).

Despite the prominence of WUI mapping, one of the most commonly reported derived quantities—WUI boundary length—is still often treated as a fixed scalar. This interpretation implicitly assumes that the boundary is uniquely defined and measured independently of analytical assumptions. Neither assumption is generally valid. Boundary length is not an intrinsic landscape constant; it is an estimand whose value depends on both boundary construction and boundary measurement.

Two distinct mechanisms generate this dependence. The first arises during boundary construction: classification choices regarding settlement representation, threshold values, neighborhood radius, and adjacency rules produce different WUI geometries (Stewart et al., 2007, Carlson et al., 2022). The second arises during boundary measurement: even for a fixed geometric object, estimated perimeter varies with measurement resolution, consistent with coastline-paradox and fractal-scaling logic (Mandelbrot, 1967, Richardson, 1961). These mechanisms are conceptually distinct but are often conflated in interpretation.

Recognizing this structure changes the scientific question. Instead of asking for one perimeter value, the relevant question is how interface geometry behaves across measurement scales under alternative delineation assumptions. Beyond measurement, the WUI boundary is best interpreted as an emergent anthropogenic–ecological interface generated by interacting spatial processes. Settlement expansion follows infrastructure networks and economic gradients, vegetation patterns emerge from ecological heterogeneity and disturbance history, and wildfire both responds to and reshapes this interface (McAlpine and Wotton, 1993, Hargrove et al., 2012, Macauley et al., 2022, Bar-Massada et al., 2014).

This study develops a unified framework for that interpretation. First, we formalize WUI boundary length as a scale-conditioned family of functions across delineation bundles and measurement scales. Second, we interpret WUI boundaries as generative anthropogenic edges produced by interacting human and ecological processes. Third, we introduce a minimal dynamical representation to clarify mechanism-level interpretation of observed boundary complexity.

2. Scale Dependence in Boundary Geometry

Irregular boundaries rarely possess a single invariant length. The coastline paradox demonstrates that measured length increases as ruler size decreases because finer measurements capture additional spatial irregularity (Mandelbrot, 1967, Richardson, 1961). In this setting, boundary length is better described as a scaling relationship than as a single scalar.

A common representation is

$$L(\varepsilon) = c\,\varepsilon^{1-D},$$

where $L(\varepsilon)$ is measured length at scale ε, c is a proportionality constant, and D is an effective roughness dimension. In logarithmic form,

$$\log L = \log c + (1 - D)\log \varepsilon,$$

which motivates slope-based inference and diagnostics over explicit scale intervals.

Wildfire perimeter studies provide direct precedent. Perimeter geometry reflects interacting controls (fuels, weather, topography), and observed fractal-like complexity changes with analysis scale (McAlpine and Wotton, 1993, Hargrove et al., 2012, Macauley et al., 2022). These findings indicate that irregular boundaries can encode process information rather than merely measurement error.

The same principle applies to WUI. Settlement structure intersects heterogeneous vegetation mosaics, producing boundaries with multi-scale irregularity. Consequently, perimeter estimates are inherently scale-conditioned and should be interpreted through curves and diagnostics rather than single values (Radeloff et al., 2005, Stewart et al., 2007, Carlson et al., 2022).

3. Definition Scale and Measurement Scale

The scale dependence of WUI boundary length arises from two independent mechanisms.

3.1. Definition Scale

Definition scale refers to assumptions embedded in delineation rules. Analysts choose settlement representation (e.g., housing-density aggregates versus building-level points), development and vegetation thresholds, neighborhood radius, adjacency criteria, and urban exclusions (Radeloff et al., 2005, Stewart et al., 2007, Carlson et al., 2022). These choices alter the mapped WUI geometry and therefore alter the boundary object itself.

3.2. Measurement Scale

Measurement scale refers to the geometric resolution used to estimate perimeter for a fixed boundary object. Smaller rulers or finer geometric approximations capture greater local sinuosity and generally return longer estimates (Mandelbrot, 1967, Richardson, 1961). Thus, perimeter variation with ε persists even when delineation assumptions are held constant.

3.3. Formal Decomposition

Let $d \in \mathcal{D}$ denote a defensible delineation bundle, and let B_d denote the resulting boundary object. Measured boundary length at geometric scale ε is

$$L_d(\varepsilon) = \mathcal{M}(B_d, \varepsilon),$$

where $\mathcal{M}$ is a specified measurement operator. Under scale-structured geometry, an interval-specific approximation is

$$L_d(\varepsilon) = c_d\, \varepsilon^{1-D_d}.$$

The estimand is therefore not a scalar but the family $\{L_d(\varepsilon) : d \in \mathcal{D}\}$. Scientifically defensible interpretation requires reporting both object variation across d and measurement variation across ε.

4. Generative Processes of the WUI Boundary

WUI boundary geometry is produced by interacting spatial processes, not by geometric partitioning alone.

Settlement expansion is typically network-mediated rather than spatially uniform. Transportation corridors, zoning regimes, economic gradients, and topographic constraints channel development into branching or clustered configurations that propagate through infill and leapfrog dynamics (Bar-Massada et al., 2014, Li et al., 2022).

Vegetation landscapes exhibit heterogeneous mosaics driven by environmental gradients, disturbance history, and ecological succession. Development advancing into these mosaics intersects patch boundaries and increases interface irregularity (Radeloff et al., 2018, Bar-Massada et al., 2014).

Wildfire introduces an additional coupling mechanism. Fire behavior responds to fuel and weather structure while also reorganizing ecological pattern through disturbance; post-fire regeneration and rebuilding can subsequently reshape boundary complexity. In this sense, wildfire both interacts with and reorganizes anthropogenic–ecological edges (Hargrove et al., 2012, Macauley et al., 2022).

The combined effect is a boundary with structure across multiple scales: infrastructure diffusion contributes branching anthropogenic form, vegetation mosaics contribute ecological irregularity, and wildfire disturbance contributes temporal reorganization of both (McAlpine and Wotton, 1993, Hargrove et al., 2012, Macauley et al., 2022, Bar-Massada et al., 2014).

5. A Minimal Dynamical Representation

To illustrate mechanism-level interpretation, consider settlement density $S(x,t)$, vegetation biomass $V(x,t)$, and wildfire activity $F(x,t)$ over spatial position x and time t.

A minimal coupled system is

$$\partial_t S = D_s \nabla^2 S + f(S, I),$$

$$\partial_t V = D_v \nabla^2 V + g(V) - h(S, V, F),$$

$$\partial_t F = k(V, W, T),$$

where D_s and D_v are diffusion-like parameters, I denotes infrastructure and development drivers, W denotes weather forcing, and T denotes topographic controls.

In this representation, WUI boundaries are contours where settlement and vegetation fields intersect under chosen thresholds. Although simplified, the system clarifies that observed interface geometry can be interpreted as the spatial trace of coupled human–environment dynamics rather than as a static cartographic artifact.

6. Implications for WUI Analysis and Decision Support

This framework clarifies several recurrent interpretation problems.

First, differences in reported WUI boundary length across studies are often expected consequences of different delineation bundles or measurement scales rather than contradictory findings. Comparability improves when studies report full d definitions, scale grids for ε, and diagnostics for the analyzed scaling interval (Bar-Massada, 2021, Hanberry, 2020).

Second, WUI boundaries should be treated as dynamic features that evolve with settlement change, ecological succession, and disturbance. Longitudinal monitoring of boundary geometry can therefore provide information about changing anthropogenic–ecological coupling (Radeloff et al., 2018, Macauley et al., 2022, Bar-Massada et al., 2023).

Third, WUI can be interpreted as one member of a broader class of anthropogenic edge systems (e.g., agricultural frontiers, forest-fragmentation edges, peri-urban transitions). This link opens methodological transfer between wildfire science and landscape-process theory (Bar-Massada et al., 2014, 2023).

For rigorous reporting, a minimum disclosure set should include: (i) complete delineation bundle d, (ii) measurement protocol and scale interval, (iii) perimeter-by-scale curves $L_d(\varepsilon)$, (iv) log–log diagnostics, and (v) sensitivity of conclusions across plausible alternative delineation bundles (Carlson et al., 2022, Bar-Massada, 2021, Hanberry, 2020).

7. Conclusions

The length of the wildland–urban interface boundary is not a fixed geographic constant but a scale-conditioned estimand. Its value depends jointly on definition scale (how the boundary is constructed) and measurement scale (how the constructed object is measured). Recognizing this structure reframes WUI perimeter estimation from a static measurement exercise into a formal inference problem.

Interpreting WUI geometry also requires a generative perspective: settlement expansion, vegetation mosaics, and wildfire disturbance jointly produce anthropogenic–ecological boundaries with multi-scale structure. Representing perimeter as $L_d(\varepsilon)$ provides a coherent framework for this interpretation, supports transparent cross-study comparison, and strengthens decision support for wildfire risk analysis, planning, insurance analytics, and policy reporting.

Author Contributions

Conceptualization: First Author and Second Author
Investigation: First Author
Writing—original draft: First Author
Writing—review and editing: First Author and Second Author

Funding

This research received no external funding.

Institutional Review Board Statement

Not applicable.

Informed Consent Statement

Not applicable.

Data Availability Statement

No new datasets were generated for this synthesis and framework study.

Acknowledgments

The authors thank collaborators for discussion and feedback.

Conflicts of Interest

The authors declare no conflict of interest.

References

Cohen, J. D. Preventing disaster: Home ignitability in the wildland–urban interface. *Journal of Forestry* **2000**, *98*, 15–21.

Radeloff, V. C.; Hammer, R. B.; Stewart, S. I.; Fried, J. S.; Holcomb, S. S.; McKeefry, J. F. The wildland–urban interface in the United States. *Ecological Applications* **2005**, *15*, 799–805. doi:10.1890/04-1413.

Stewart, S. I.; Radeloff, V. C.; Hammer, R. B.; Hawbaker, T. J. Defining the wildland–urban interface. *Journal of Forestry* **2007**, *105*, 201–207. doi:10.1093/jof/105.4.201.

Carlson, A. R.; Helmers, D. P.; Hawbaker, T. J.; Mockrin, M. H.; Radeloff, V. C. The wildland–urban interface in the United States based on 125 million building locations. *Ecological Applications* **2022**, e2597. doi:10.1002/eap.2597.

Radeloff, V. C.; Helmers, D. P.; Kramer, H. A.; Mockrin, M. H.; Alexandre, P. M.; Bar-Massada, A.; et al. Rapid growth of the U.S. wildland–urban interface raises wildfire risk. *Proceedings of the National Academy of Sciences* **2018**, *115*, 3314–3319. doi:10.1073/pnas.1718850115.

Mandelbrot, B. B. How long is the coast of Britain? Statistical self-similarity and fractional dimension. *Science* **1967**, *156*, 636–638. doi:10.1126/science.156.3775.636.

Richardson, L. F. The problem of contiguity: An appendix to statistics of deadly quarrels. *General Systems Yearbook* **1961**, *6*, 139–187.

McAlpine, R. S.; Wotton, B. M. The use of fractal dimension to improve wildland fire perimeter predictions. *Canadian Journal of Forest Research* **1993**, *23*, 1073–1077. doi:10.1139/x93-137.

Hargrove, W. W.; Hoffman, F. M.; Efroymson, R. A.; Stockton, T.; Shugart, H. H. A geometry-based method for fractal estimation of wildfire perimeters. *PLOS ONE* **2012**, *7*, e38232. doi:10.1371/journal.pone.0038232.

Macauley, K. A. P.; McLoughlin, N.; Beverly, J. L. Modelling fire perimeter formation in the Canadian Rocky Mountains. *Forest Ecology and Management* **2022**, *506*, 119958. doi:10.1016/j.foreco.2021.119958.

Bar-Massada, A.; Radeloff, V. C.; Stewart, S. I. Biotic and abiotic effects of human settlements in the wildland–urban interface. *BioScience* **2014**, *64*, 429–437. doi:10.1093/biosci/biu036.

Calkin, D. E.; Cohen, J. D.; Finney, M. A.; Thompson, M. P. How risk management can prevent future wildfire disasters in the wildland–urban interface. *Proceedings of the National Academy of Sciences* **2014**, *111*, 746–751. doi:10.1073/pnas.1315088111.

Li, S.; Dao, V.; Kumar, M.; Nguyen, P.; Banerjee, T. Mapping the wildland–urban interface in California using remote sensing data. *Scientific Reports* **2022**, *12*, 5789. doi:10.1038/s41598-022-09707-7.

Bar-Massada, A. A comparative analysis of two major approaches for mapping the wildland–urban interface: A case study in California. *Land* **2021**, *10*, 679. doi:10.3390/land10070679.

Hanberry, B. B. Reclassifying the wildland–urban interface using fire occurrences for the United States. *Land* **2020**, *9*, 225. doi:10.3390/land9070225.

Bar-Massada, A.; Alcasena, F.; Schug, F.; Radeloff, V. C. The wildland–urban interface in Europe: Spatial patterns and associations with socioeconomic and demographic variables. *Landscape and Urban Planning* **2023**, *235*, 104759. doi:10.1016/j.landurbplan.2023.104759.